Open Source Software Engineering: An Introduction to Open Source Tools

Software development is one of the youngest branches of engineering. It is the study and application of engineering principles and methodologies to software development, with the aim of producing quality software products. In this series, the author will explore specific open source tools that are relevant to software engineering.

A long time back, I remember reading a popular white paper on software engineering. It said that software engineers could be compared to a cobbler's barefoot children: *"They make tools and applications that enable users in many domains to perform their work more effectively and efficiently, yet frequently, they do not use those tools themselves."* To some extent this remains true even today, because software product engineering still requires a lot of support, from the tools point of view. Thanks to open source, we not only get the source code for development, but also get a bunch of tools to deliver high quality products.

The relevance of open source software engineering

Now, one may ask, "Why do we need to have such tools? What is the importance of following software engineering processes?" Well, the answer is simple. Based on experience, we know that building a product inside a lab and getting a momentary high is quite different from deploying a commercial quality product to a customer who pays for it. In order to achieve the later, project managers, product engineers, architects and quality professionals face multiple challenges. To a larger extent, these challenges are overcome by implementing various processes and adopting different lifecycles like Waterfall, Agile, etc.

In each of these lifecycles, there are engineering activities that are defined, like requirement analysis, design, coding, customer demos, etc. These engineering activities help software teams to set up activities that are benchmarks and repeatable. This eventually builds quality in each cycle and ensures predictability in software delivery. Quality needs to be controlled and managed throughout the software development lifecycle, or it could lead to customer dissatisfaction or even major disasters, when projects fail on a large scale. The schedule is another critical element that needs to be managed throughout the software development process.

There are innumerable organisations that have released various expensive tools to manage software products, but these are beyond the budget of start-ups and entrepreneurial ventures. Due to constantly